

Lecture 6: Stocks, Stock Markets, and Market Efficiency

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Overview

For individuals, stocks are a key instrument for holding wealth; for companies, they are one of several ways to obtain financing.

Key insight:

- Stock markets are central links between the financial world and the real economy
- Understanding stock valuation is essential for investors and policymakers
- Efficient markets theory helps explain stock price behavior

The Road Ahead

1. Essential Characteristics of Stock
2. Valuing Stocks
3. Market Efficiency

Essential Characteristics of Stock

Stocks, also known as equity, are shares in a firm's ownership. When a firm issues stock, it sells part of itself so that the buyer becomes a part owner.

Three characteristics:

1. Small denominations: Shares are issued in small units, allowing investors to buy as little or as much of the company as they want
2. Transferable: Owners can sell their shares to someone else
3. Residual claimants with limited liability: Stockholders receive what is left after all other creditors are paid, but the maximum they can lose is their initial investment if the company goes bankrupt

Measuring Stock Market

Changes in stock values affect consumption and saving patterns, causing general economic activity to fluctuate.

Index	Companies	Weighting	Key Feature
DJIA	30 largest	Price-weighted	Higher-priced stocks dominate
S&P 500	500 largest	Value-weighted	Higher market value stocks dominate
Nasdaq	~3,000	Value-weighted	Heavy tech representation
Wilshire 5000	All publicly traded	Value-weighted	Most comprehensive

Valuing Stocks

The fundamental value of a firm's stock depends on its current assets and estimates of future profitability.

Key question: What determines the price of a stock?

We will develop the **dividend-discount model** to answer this question by:

1. Starting with a simple one-year investment
2. Extending to multiple periods
3. Incorporating dividend growth
4. Adding risk considerations

One-Year Investment

Suppose an investor plans to buy a stock today and sell it in one year.

Notation:

- P_t = purchase price today
- P_{t+1} = sale price one year later
- D_{t+1} = dividend payment next year
- i = interest rate for computing present value

Stock price today:

$$P_t = \frac{D_{t+1}}{1+i} + \frac{P_{t+1}}{1+i}$$

Multi-Period Extension

Similarly, the stock price next year is:

$$P_{t+1} = \frac{D_{t+2}}{1+i} + \frac{P_{t+2}}{1+i}$$

Substituting this into the previous equation:

$$P_t = \frac{D_{t+1}}{1+i} + \frac{D_{t+2}}{(1+i)^2} + \frac{P_{t+2}}{(1+i)^2}$$

Repeating this process for n periods:

$$P_t = \frac{D_{t+1}}{1+i} + \frac{D_{t+2}}{(1+i)^2} + \dots + \frac{D_{t+n}}{(1+i)^n} + \frac{P_{t+n}}{(1+i)^n}$$

Interpretation: Stock price today is the present value of all future dividends plus the present value of the sale price n years from now.

Dividend-Discount Model

Assume dividends grow at a constant rate g per year:

$$D_{t+n} = D_t(1 + g)^n$$

If the firm pays dividends forever (i.e., $n \rightarrow \infty$), the stock price becomes:

$$P_t = \frac{D_t(1 + g)}{i - g} \quad \text{for } i > g$$

Key insight: Stock prices should be high when:

- Current dividends (D_t) are high
- Dividend growth (g) is rapid
- Interest rate (i) is low

Why Stocks Are Risky

Leverage is the practice of borrowing to finance part of an investment, and it creates risk.

Example: A software company needs a \$1,000 computer financed by stock/bonds (10% rate). Earnings: \$160 (good years) or \$80 (bad years) with equal probability.

Table 8.3

Returns Distributed to Debt and Equity Holders under Different Financing Assumptions

Percent Equity (%)	Percent Debt (%)	Required Payments on 10% Bonds (\$)	Payment to Equity Holders (\$)	Equity Return (%)	Expected Equity Return (%)	Standard Deviation of Equity Return
100%	0	0	\$80–\$160	8–16%	12%	4%
50%	50%	\$50	\$30–\$110	6–22%	14%	8%
30%	70%	\$70	\$10–\$90	$3\frac{1}{3}$ –30%	$16\frac{2}{3}$ %	$13\frac{1}{3}$ %
20%	80%	\$80	\$0–\$80	0–40%	20%	20%

Key insight: The more debt, the more leverage, and the greater the stockholders' risk because they are residual claimants.

Risk and the Value of Stocks

Stockholders require compensation for bearing risk. The higher the risk, the greater the compensation required.

Dividend-discount model with risk:

$$P_t = \frac{D_t(1 + g)}{rf + rp - g}$$

where:

- rf = risk-free interest rate
- rp = equity risk premium

Key insight: Stock prices are higher when [i.] current dividends are high, [ii.] dividends are expected to grow quickly, [iii.] risk-free rate is low, and [iv.] equity risk premium is low.

Market Efficiency

The theory of efficient markets states that prices of all financial instruments reflect all available information.

Key implications:

1. Markets adjust immediately and continuously to changes in fundamental values
2. Stock price movements are unpredictable because they only respond to new information, which by definition is unpredictable
3. No one can consistently beat the market: If prices already reflect all available information, no investor can systematically earn above-average returns

Grossman-Stiglitz Paradox

Grossman and Stiglitz (1980) identified a fundamental problem with perfectly efficient markets: the impossibility of informationally efficient markets.

Paradox:

1. If markets were perfectly efficient, prices would fully reflect all information, so no one could profit from costly information gathering
2. Without profit opportunities, no one would have incentive to acquire information
3. But if no one gathers information, how can prices reflect information?

Resolution: Markets must be only partially efficient in equilibrium, providing just enough profit opportunities to compensate informed traders for their information costs.